Langchain Demo With LLAMA2 API

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write a binary search tree code

Sure! Here is an example implementation of a binary search tree in Python:

```
class BinarySearchTree:
    def __init__(self):
        self.root = None

    def insert(self, value):
        if not self.root:
            self.root = Node(value)
        else:
            self.insert_node(self.root, value)

    def insert_node(self, node, value):
        if value < node.value:
            if not node.left:
                node.left = Node(value)
            else:
                self.insert_node(node.left, value)
        elif value > node.value:
            if not node.right:
                node.right = Node(value)
            else:
                self.insert_node(node.right, value)
        else:
            raise ValueError("Value already exists in the tree")

    def search(self, value):
        current = self.root
        while current:
            if value == current.value:
                return current
            elif value < current.value:
                current = current.left
            else:
```

```
                current = current.right
        return None

    def traverse(self, node):
        if node:
            self.traverse(node.left)
            print(f"{node.value}: {node.left}")
            self.traverse(node.right)
            print(f"{node.value}: {node.right}")

class Node:
    def __init__(self, value):
        self.value = value
        self.left = None
        self.right = None
```

This implementation defines a `BinarySearchTree` class with methods for inserting nodes, searching for a specific value, and traversing the tree in a depth-first manner. The `Node` class represents a single node in the tree, with fields for the value and references to the left and right child nodes.

To use this implementation, you can create a new instance of the `BinarySearchTree` class and insert nodes using the `insert` method. You can search for a specific value using the `search` method, and traverse the tree using the `traverse` method.

Here's an example usage of the `BinarySearchTree` class:

```
# Create a new binary search tree
bt = BinarySearchTree()

# Insert some nodes
bt.insert(5)
bt.insert(2)
bt.insert(8)
bt.insert(3)

# Search for a specific value
print(bt.search(5)) # Should print 5
print(bt.search(2)) # Should print 2
print(bt.search(8)) # Should print 8
print(bt.search(10)) # Should raise ValueError
```

```
# Traverse the tree
bt.traverse(bt.root)
```

This code creates a new `BinarySearchTree` instance, inserts some nodes into it, searches for a specific value, and traverses the tree in a depth-first manner.